

ENGR 225 Homework 3 Questions

Assoc. Prof. Clayton Byers

Assigned: 25 Sept 2026

Due: 2 Oct 2026 (11:59pm on Moodle)

Important notes before you begin:

- I've been asking you to draw your analysis or sketch a setup - now that turns into drawing the Free Body Diagram (FBD) for your problems! You should be doing this for every problem you encounter where you balance forces and moments. (Problems 4 and onward in this assignment, but don't neglect the analysis of the first three!)
- Remember to submit the cover sheet with your assignment. Consolidate all your work to a single document.

Problem 1

Around 85% of all neck injuries are due to rear-end car collisions. The design of a seat restraint aims to reduce the number of injuries by providing a distributed pressure with the head during impact. The load applied is measured to be parabolic as shown in the figure below. Determine the equivalent resultant force and its location measured from point A. (**Ans:** $F = 7\text{ lb}$ and $\bar{x} = 0.268\text{ ft}$)

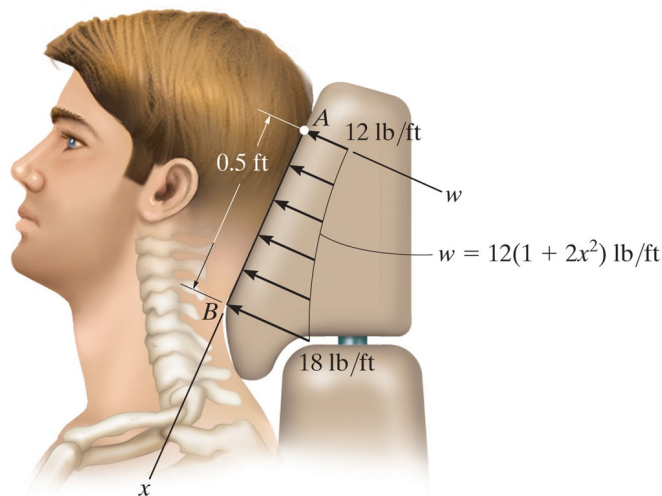


Figure for problem 1.

Problem 2

Replace system of loads on the beam by an equivalent force and couple moment acting at the point O . (**Ans:** $F_R = -51 \text{ kN}$ and $M_R = -914 \text{ kN}$)

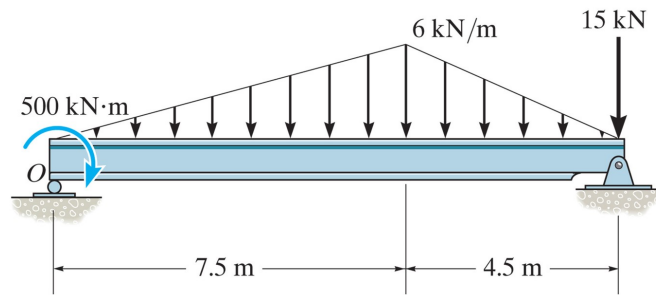


Figure for problem 2.

Problem 3

Replace the distributed load by an equivalent resultant force and specify where its line of action intersects a horizontal line along member AB , measured from A . (**Ans:** $\vec{R} = -8\hat{i} - 9\hat{j} \text{ kN}$ and $d = 3.28 \text{ m}$)

Note: If you are replacing all the loads with a single force, how does that play into the equivalent moment?

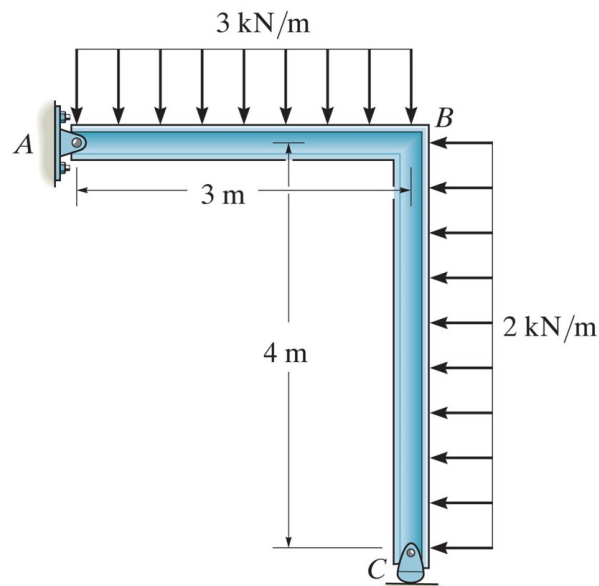


Figure for problem 3.

Problem 4

Determine the maximum force $\vec{F}$ that can be supported in the position shown if each chain can support a maximum tension of 600 lb before it fails. (**Ans:** $F = 960\text{ lb}$)

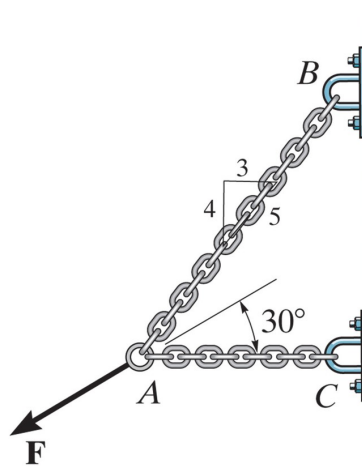


Figure for problem 4.

Problem 5

Determine the unstretched length of DB to hold the 40 kg crate in the position shown. Set $k = 180\text{ N/m}$. (**Ans:** $\ell_0 = 2.03\text{ m}$)

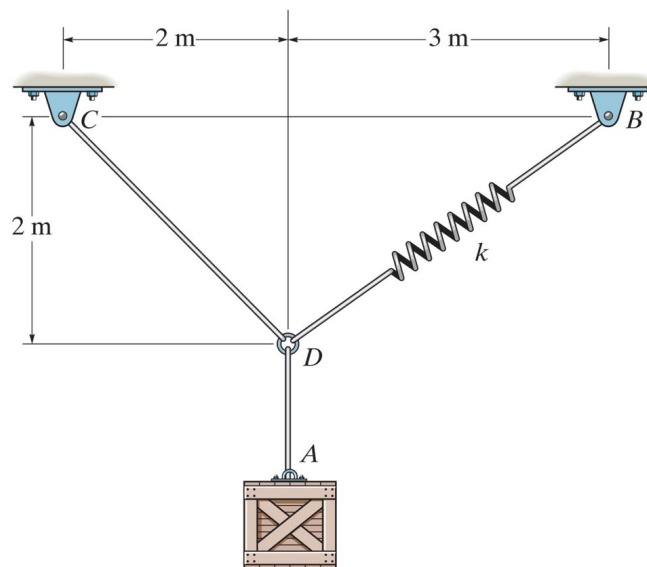


Figure for problem 5.

Problem 6

Each cord can sustain a maximum tension of 500 N . Determine the largest mass that can be supported at the hook H . (**Ans:** $m = 26.7\text{ kg}$)

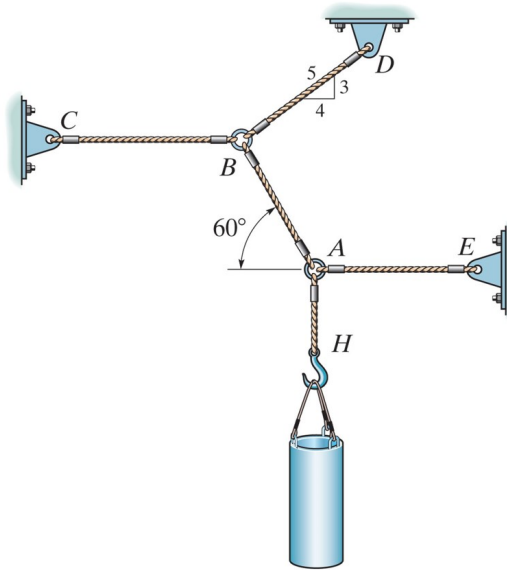


Figure for problem 6.

Problem 7

Determine the stretch in each of the two springs required to hold the 20 kg crate in equilibrium as shown below. Each spring has an unstretched length of 2 m and a stiffness of $k = 300\text{ N/m}$. (**Ans:** $s_{OA} = 0.218\text{ m}$ and $s_{OB} = 0.327\text{ m}$)

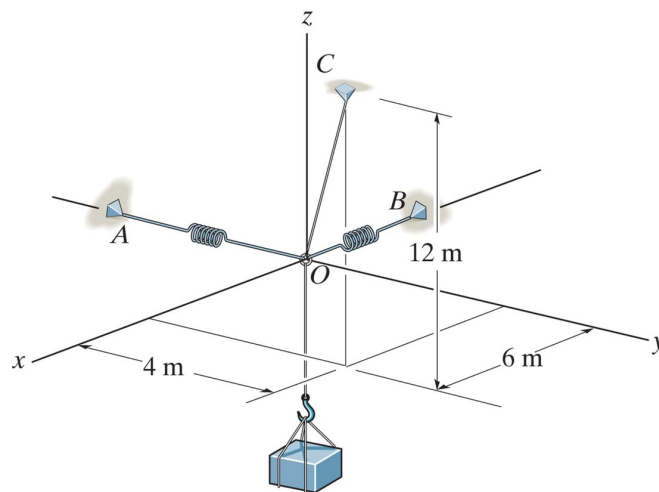


Figure for problem 7.

Problem 8

The 800 lb cylinder is supported by three chains as shown below. Determine the force in each chain needed to maintain equilibrium. Set $d = 1\text{ ft}$. (**Ans:** $F_{AB} = 469\text{ lb}$, $F_{AC} = F_{AD} = 331\text{ lb}$)

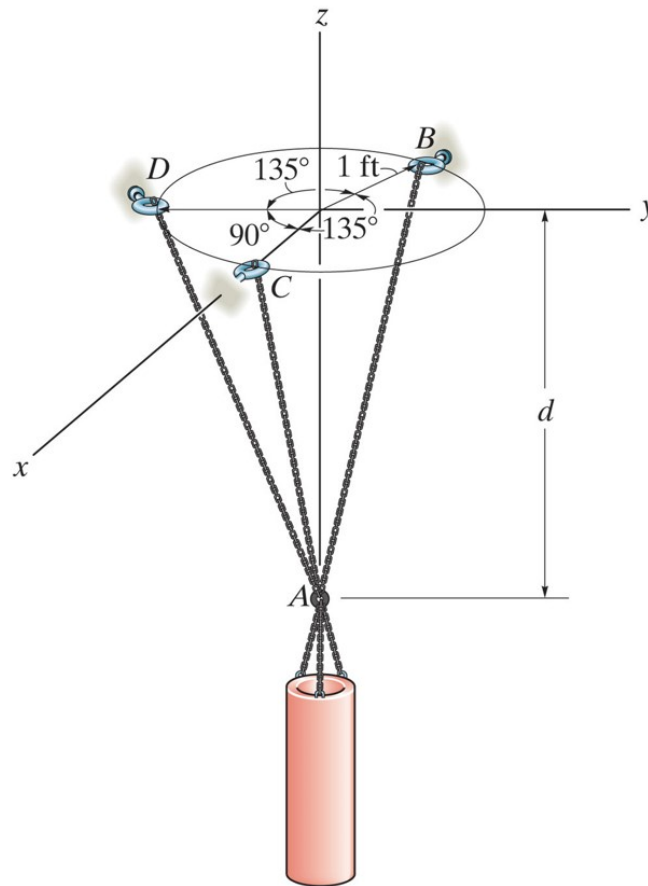


Figure for problem 8.